Electric Vehicles Data Analysis – Final Report

1. Executive Summary

This dashboard provides a comprehensive analysis of Electric Vehicle (EV) adoption across years, states, manufacturers, and vehicle types. The insights reveal strong market growth, dominance of specific players, and the critical role of policy and infrastructure in shaping EV adoption.

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2. Key KPIs Overview

- **Total Vehicles:** 1,50,437
- **Average Electric Range:** 67.86 km
- **BEV Vehicles:** 1,16,766 (**77.6%**)
- **PHEV Vehicles:** 33,671 (**22.4%**)

👉 **Insight:** The EV market is heavily dominated by **Battery Electric Vehicles (BEVs)**, indicating a clear shift toward full electrification.

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3. Trend Analysis (Vehicles by Model Year)

📈 **Key Observations:**

- EV adoption remained slow until **2015**, then gradually increased
- Strong growth from **2018 onwards**
- Peak in **2023 (~37K vehicles)**
- Sharp drop in **2024 (~0.6K)** (likely incomplete data)

👉 **Insight:**

- The market is in a **rapid expansion phase**
- Growth pattern is **exponential, not linear**
- 2024 data should be treated as **partial/incomplete**

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4. Geographic Analysis (Vehicles by State)

🌍 **Key Observations:**

- **California (92K vehicles)** dominates EV adoption
- Other states show significantly lower counts

- Adoption is concentrated in **few high-performing states**

👉 **Insight:**

- EV adoption is **regionally concentrated**
- High adoption correlates with:
  - Strong infrastructure
  - Government incentives
  - Urbanization

👉 **Business Meaning:**
There is a **huge untapped market** in low-adoption states.

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5. Manufacturer Analysis (Top 10 Makes)

🏭 **Key Observations:**

- **Tesla alone holds ~52.7% market share (68,960 vehicles)**
- Followed by:
  - Nissan (~10.3%)
  - Chevrolet (~9.2%)
- Remaining players have **very small shares**

👉 **Insight:**

- EV market shows **high market concentration**
- Tesla is the **clear market leader**

👉 **Risk:**

- Market dependency on a single player

👉 **Opportunity:**

- Space for **new competitors and innovation**

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6. Model-Level Insights

🚙 **Top Models:**

- **Tesla Model Y:** 28,501 (18.95%)
- **Tesla Model 3:** 27,708 (18.42%)
- **Nissan Leaf:** 13,187 (8.77%)

👉 **Insight:**

- Tesla dominates not just at brand level, but also at **model level**
- Few models contribute a **large portion of total EV adoption**

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7. CAFV Eligibility Analysis

📊 **Distribution:**

- CAFV Eligible: **41.82%**
- CAFV Not Eligible: **11.85%**
- Unknown: **46.33%**

👉 **Insight:**

- Nearly **half of the data is unknown**, indicating data quality issues
- Still, a large portion benefits from **government incentives**

👉 **Business Meaning:** Policies significantly influence EV adoption.

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8. Vehicle Type Analysis

🚙 **BEV vs PHEV:**

- BEV: **77.6%**
- PHEV: **22.4%**

👉 **Insight:**

- Consumers prefer **fully electric vehicles over hybrids**
- Indicates growing confidence in:
    - Battery technology
    - Charging infrastructure

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9. Electric Range Insight

- Average range: **67.86 km**

👉 **Insight:**

- Moderate range suggests:
  - Many vehicles are older or entry-level
- Range is still a **key decision factor**

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10. Hidden Insights (Advanced Level)

🔍 **1. Market Maturity Stage**

- Data suggests transition from:
  Early Adoption → Early Majority

🔍 **2. Infrastructure Dependency**

- States with high EV adoption likely have:
  - Better charging networks
  - Strong policy support

🔍 **3. Data Quality Issue**

- High "CAFV Unknown" → needs cleaning
  👉 Important for real-world analytics

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11. Business Recommendations

🏛 **For Government:**

- Expand EV infrastructure in low-adoption states
- Improve data tracking (reduce "unknown" category)
- Provide targeted subsidies

🏭 **For Manufacturers:**

- Focus on affordable EVs
- Improve battery range
- Expand beyond saturated markets

💰 **For Investors:**

- Invest in:
    - Charging infrastructure
    - Battery technology
    - Emerging EV markets

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12. Conclusion

The EV market is experiencing **rapid growth**, led by dominant players like Tesla and driven by government incentives and infrastructure availability. However, adoption remains uneven, and significant opportunities exist in underserved regions.

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